

Mechanism of action of stinging nettles.



OBJECTIVE: Inadvertent exposure to the ubiquitous weed, *Urtica dioica*, called "stinging nettles" produces an immediate stinging and burning sensation on the skin. This investigation evaluates the structural effect that stinging nettle spicules may have on the clinical manifestation of these symptoms. This hypothesis was investigated by exposing murine skin to stinging nettles and then evaluating the skin using electron microscopy. It was hypothesized that the mechanism of action of stinging nettles is both biochemical and mechanical, which may have clinical significance regarding treatment for acute exposure. **METHODS:** Fresh post-mortem dermis samples from the carcasses of genetically modified hairless mice were brushed under the stem and leaf of a stinging nettle plant, mimicking the clinical method of exposure a patient might experience. Another set of mouse skin samples was obtained but not exposed to the nettles. Both sets of skin samples were imaged with scanning electron microscopy. **RESULTS:** The skin samples that were not exposed to nettle leaves were uniform, with occasional striated hairs on the skin surface and no nettle spicules. The skin samples exposed to nettle leaves showed many smooth nettle spicules piercing the skin surface. A few spicules retained their bases, which appear empty of any liquid contents. **CONCLUSIONS:** The mechanism of action of stinging nettles dermatitis appears to be both biochemical and mechanical. Impalement of spicules into the skin likely accounts for the mechanical irritation in addition to the known adverse chemical effects of stinging nettles. Further investigation of treatment modalities is warranted. Copyright © 2011 Wilderness Medical Society. Published by Elsevier Inc. All rights reserved.